

Question Number	Answer	Additional Guidance	Mark
4(a)	An answer that includes three of the following: • find a suitable method of identifying this species (1) • find a suitable method to stimulate grasshopper movement (because they are camouflaged) (1) • find the time of day/month the grasshoppers are (most) active (1) • find a suitable {sampling/counting} method (1) • find suitable weather conditions (1)	Accept the best time of day/month to do the study E.g. suitable methods - transects/size of quadrat/capture and release/nets /how to measure the grasshopper population	(3)

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4(b)	An answer that includes eight of the following: • clear statement of the dependent variable (1) • method of producing standardised sample sites (of known area on both sites) (1) • standardised {method/time} of counting on both sites (1) • identify two variables to be monitored (1) • Describe how one variable can be {monitored/controlled} (1) • repeats for {one/either} sample area (1) • repeat the whole investigation at different times of year (between November and March) (1) • method of calculation of population density (1)	Eg number of grasshoppers per unit area Allow use of nets/quadrats/camera suitable way to mark/recapture grasshoppers /Lincoln index Accept control Eg temperature, light(intensity) , humidity, rainfall, pH, time of day, weather use similar aspect/slope of road and gravel area Accept different days	(8)

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4(c)	- raw data table with headings and appropriate units, with means calculated from repeats (1) - bar graph with labelled axes (1) - use of an appropriate statistical test for difference (1)	Accept any headers with units Accept description of calculating mean Do not accept units in the body of the table Accept labels on graph from table	(3)
Question Number	Answer	Additional Guidance	Mark
4(d)	An answer that includes three of the following: - difficulty in identifying this species (1) - difficulty in seeing/making grasshopper move. (1) - difficult to ensure each grasshopper counted is a new individual (1) - difficult to {measure/monitor} a named variable (1) - Sampled once so population may change over time	Accept rare/hard to find Accept hard to catch them Accept cannot/difficult to control Accept can only be sampled once as they are endangered	(3)

(Total for question 4 = 17 mark)

